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Query 1:

Return the names and phone numbers of all MPs with no Twitter information. The result-set must contain name and phonenumber attributes, ordered by ascending name, and contain a maximum of five tuples.

SQL query:

```
select name, phonenumber
from mp
where twitter is null or twitter = ''
order by name asc
limit 5;
```

This query returns this data set

	name	phonenumber
1	Dr Anne Aly MP	(08) 9409 4517
2	Dr Mike Freelander MP	(02) 4620 0293
3	Hon Brendan O'Connor MP	(03) 8390 6166
4	Hon Craig Laundy MP	(02) 9715 7444
5	Hon Damian Drum MP	(03) 5821 5371

Query 2:

Which MPs are also a “Doctor”?. Define “Doctor” as “name starts with “’Dr”. The result set must only contain the name attribute, ordered by ascending name, and contain a maximum of five tuples.

SQL query:

```
select name
from mp
where name like 'Dr%'
order by name asc
limit 5;
```

The result:

	name
1	Dr Anne Aly MP
2	Dr Jim Chalmers MP
3	Dr Mike Freelander MP

Query 3

Some context for this query. When I imported table 2 into the db, I changed the data type of table 2 column area_albers_sqkm to numeric. This will configure the query to return a cleaner looking number when the sum() function is used

to: Column	to: Type	from: Column
SA1_MAINCODE_2016	text	SA1_MAINCODE_2016
CED_CODE_2017	integer	<Auto> (CED_CODE_2017)
CED_NAME_2017	text	<Auto> (CED_NAME_2017)
STATE_CODE_2017	integer	<Auto> (STATE_CODE_2017)
STATE_NAME_2017	text	<Auto> (STATE_NAME_2017)
AREA_ALBERS_SQKM	numeric	AREA_ALBERS_SQKM

Query: **What is the total area in square kilometers per state? The result set must contain state name and sum attributes, ordered by ascending state name, and contain all states in the data-set.**

SQL query:

```
select state_name_2017 as state_name,
sum(area_albers_sqkm) as sum
from "2"
group by state_name_2017
order by state_name asc
```

The result:

	state_name	sum
1	Australian Capital Territory	2358.172
2	New South Wales	800810.7823
3	Northern Territory	1348094.3281
4	Other Territories	256.3162
5	Queensland	1730172.0833
6	South Australia	984274.908
7	Tasmania	68017.8607
8	Victoria	227495.6304
9	Western Australia	2526646.2389

Query 4

For each party, return the sum of the district populations, but only if the party's MP is on Twitter. The result must contain all parties ordered by ascending party. The result set must contain party and sum attributes, ordered by ascending party, and contain all parties in the data-set.

SQL Query:

```
with twitter_mps as (
   select
       party,
       split_part(district, ', ', 1) as ced_name,
       split_part(district, ', ', 2) as state_name
   from mp
   where twitter is not null and twitter <> ''
),
    sa1_to_ced as(
        select
            sa1_maincode_2016,
            ced_name_2017,
            state_name_2017
        from "2"
    )
select
   m.party,
   SUM(c.tot_p_p) as sum
from twitter_mps m
        join sa1_to_ced g
             on g.ced_name_2017 = m.ced_name
                 and g.state_name_2017 = m.state_name
        join "1" c
             on c.sa1_7digitcode_2016 =
substring(g.sa1_maincode_2016::text, 5, 7)::int
group by m.party
order by m.party asc
```

	party	sum
1	Australian Greens	47431
2	Australian Labor Party	4414595
3	Independent	210161

The result:

Appendix: Setup and Data Preprocessing and Normalization

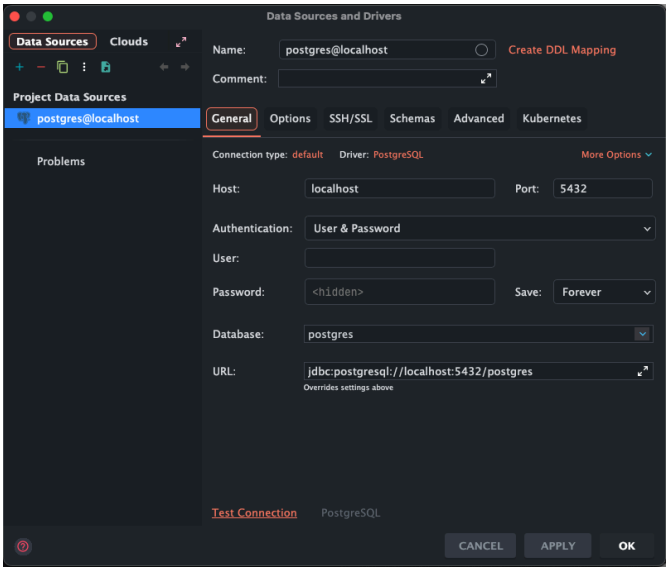
First I started Postgres on my mac

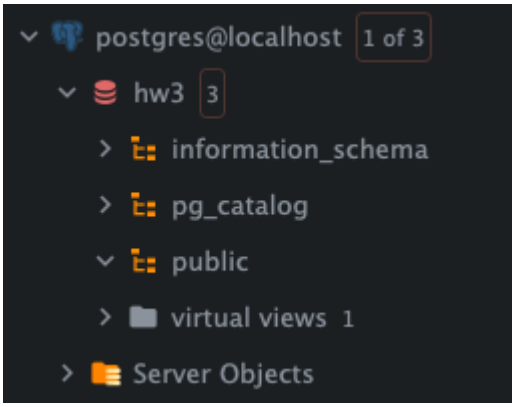
```
yaakov — -zsh — 80×24
yaakov@Yaakovs-MacBook-Pro ~ % brew services restart postgresql@14
Stopping `postgresql@14`... (might take a while)
==> Successfully stopped `postgresql@14` (label: homebrew.mxcl.postgresql@14)
==> Successfully started `postgresql@14` (label: homebrew.mxcl.postgresql@14)
yaakov@Yaakovs-MacBook-Pro ~ %
```

On the command line I run the command “psql postgres” to open the postgres console and add the database “hw3” and confirm with \l. We see the db hw3

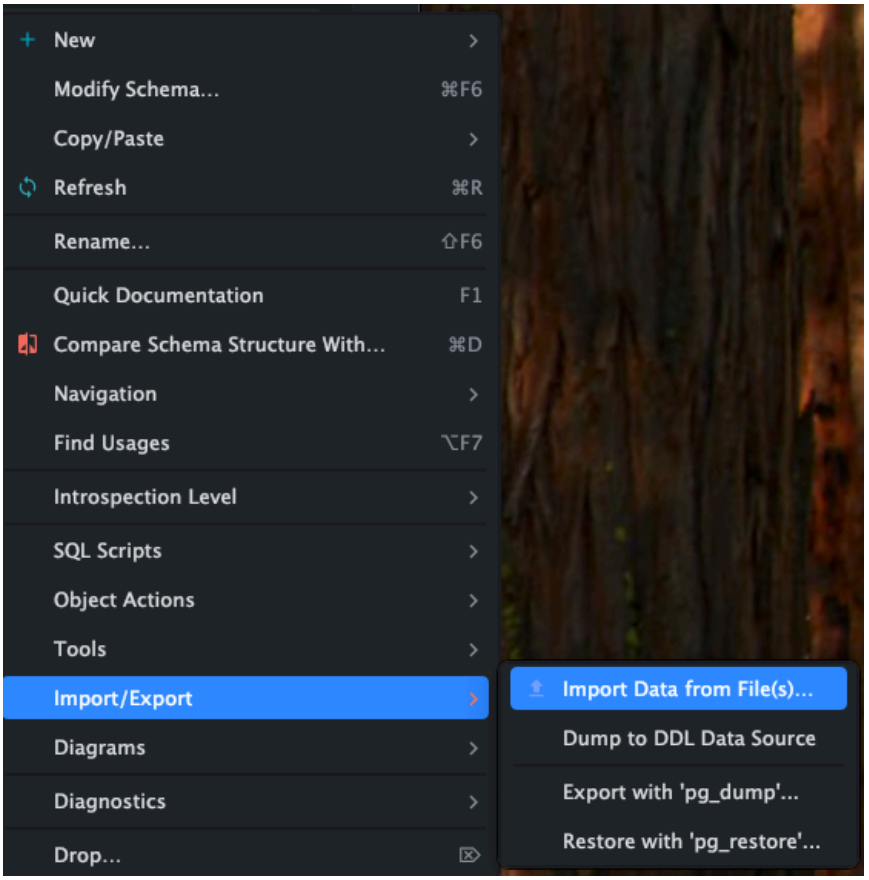
```
yaakov — psql postgres — 80×24
yaakov@Yaakovs-MacBook-Pro ~ % psql postgres
psql (14.20 (Homebrew))
Type "help" for help.

postgres=# create database hw3
postgres-# \l
                                  List of databases
   Name    | Owner  | Encoding |   Collate   |    Ctype    | Access privileges
-----------+--------+----------+-------------+-------------+-------------------
 hw3       | yaakov | UTF8     | en_US.UTF-8 | en_US.UTF-8 |
 postgres  | yaakov | UTF8     | en_US.UTF-8 | en_US.UTF-8 |
 sql10     | yaakov | UTF8     | en_US.UTF-8 | en_US.UTF-8 |
 template0 | yaakov | UTF8     | en_US.UTF-8 | en_US.UTF-8 | =c/yaakov        +
           |        |          |             |             | yaakov=CTc/yaakov
 template1 | yaakov | UTF8     | en_US.UTF-8 | en_US.UTF-8 | =c/yaakov        +
           |        |          |             |             | yaakov=CTc/yaakov
(5 rows)

postgres-#
```

Then I connected Postgres to DataGrip IDE and we can confirm that we are connected

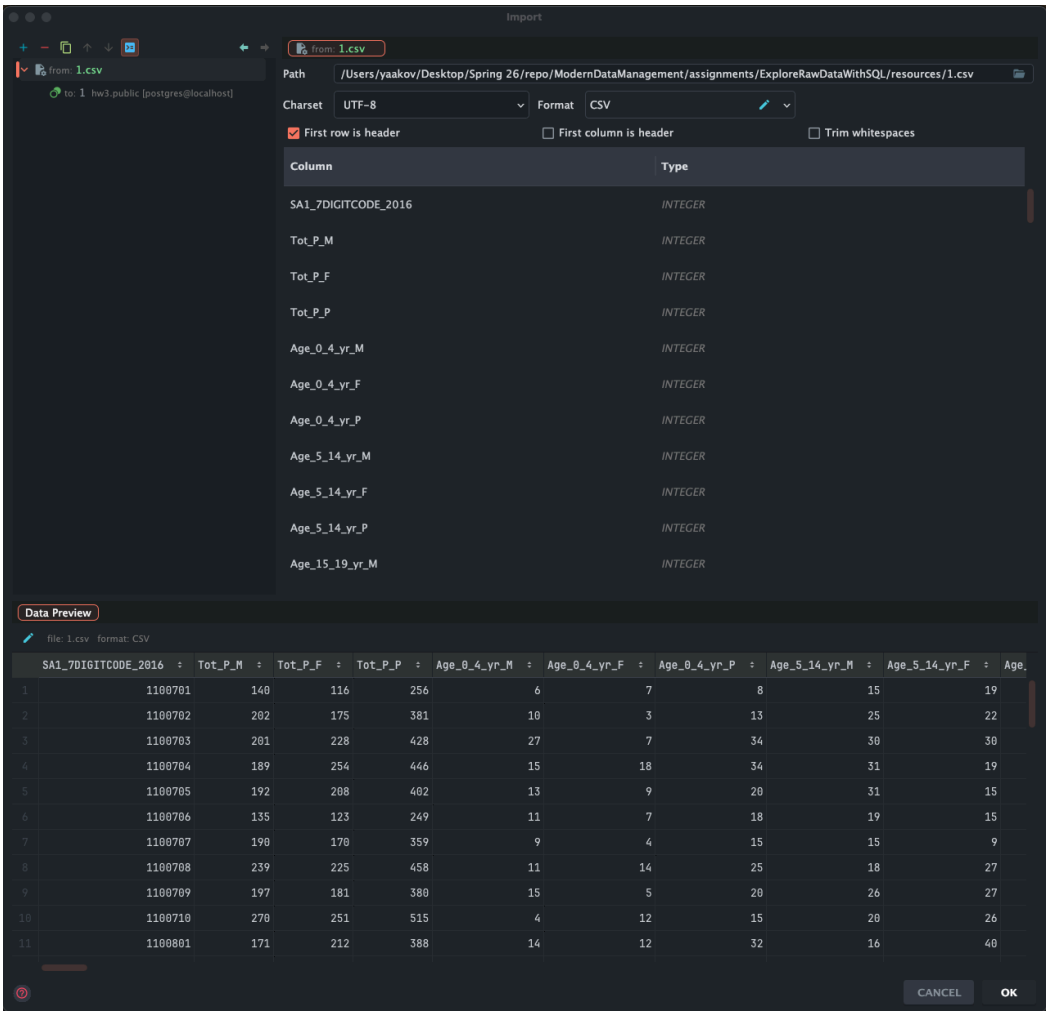




We can now import our csv files by right clicking public>import/export>import Data from file

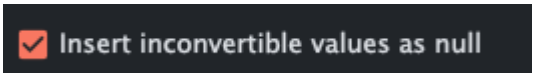


We choose a csv file and a window pops up with our formatted data turned into a table.

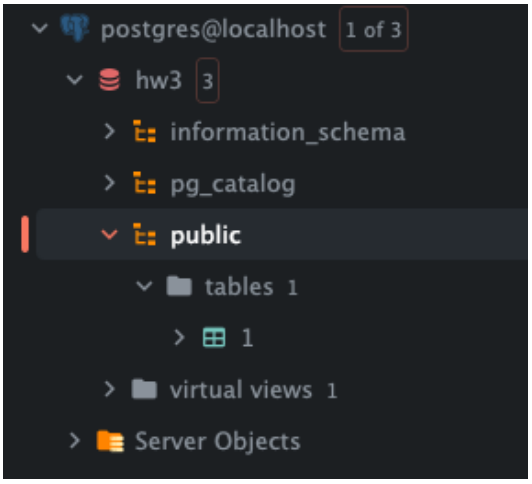


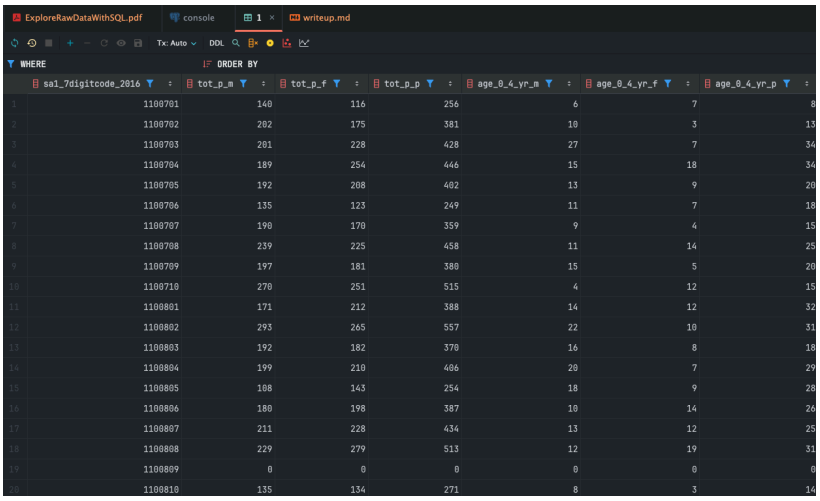
SA1_7DIGITCODE_2016	Tot_P_M	Tot_P_F	Tot_P_P	Age_0_4_yr_M	Age_0_4_yr_F	Age_0_4_yr_P	Age_5_14_yr_M	Age_5_14_yr_F	Age_
1100701	140	116	256	6	7	8	15	19	
1100702	202	175	381	10	3	13	25	22	
1100703	201	228	428	27	7	34	30	30	
1100704	189	254	446	15	18	34	31	19	
1100705	192	208	402	13	9	20	31	15	
1100706	135	123	249	11	7	18	19	15	
1100707	190	170	359	9	4	15	15	9	
1100708	239	225	458	11	14	25	18	27	
1100709	197	181	380	15	5	20	26	27	
1100710	270	251	515	4	12	15	20	26	
1100801	171	212	388	14	12	32	16	40	

We select the first row as header for csv files that have the first row as the column names. And we set “Insert inconvertible values as null”



We click ok and in our database view we now have the table





	sa1_7digitcode_2016	tot_p_m	tot_p_f	tot_p_p	age_0_4_yr_m	age_0_4_yr_f	age_0_4_yr_p
1	1100701	140	116	256	6	7	8
2	1100702	202	175	381	10	3	13
3	1100703	201	228	428	27	7	34
4	1100704	189	254	446	15	18	34
5	1100705	192	208	402	13	9	20
6	1100706	135	123	249	11	7	18
7	1100707	190	170	359	9	4	15
8	1100708	239	225	458	11	14	25
9	1100709	197	181	380	15	5	20
10	1100710	270	251	515	4	12	15
11	1100801	171	212	388	14	12	32
12	1100802	293	265	557	22	10	31
13	1100803	192	182	370	16	8	18
14	1100804	199	210	406	20	7	29
15	1100805	108	143	254	18	9	28
16	1100806	180	198	387	10	14	26
17	1100807	211	228	434	13	12	25
18	1100808	229	279	513	12	19	31
19	1100809	0	0	0	0	0	0
20	1100810	135	134	271	8	3	14

We can double click on 1 to open a view of the table. (^table1)

We can now do this to the other csv files. 1.csv becomes table 1, 2.csv becomes table 2 and 3.csv becomes table mp

ExploreRawDataWithSQL.pdf | console | mp | 1

Tx: Auto DDL

WHERE ORDER BY

	district	link	name	party	phonenum…	twitter
1	Warringah, New S…	https://www.ap…	Hon Tony Abbo…	Liberal Party…	(02) 9977 6411	http://twitter.com/…
2	Bennelong, New S…	https://www.ap…	Mr John Alexa…	Liberal Party…	(02) 9869 4288	http://twitter.com/…
3	Grayndler, New S…	https://www.ap…	Hon Anthony A…	Australian La…	(02) 9564 3588	http://twitter.com/…
4	Mallee, Victoria	https://www.ap…	Mr Andrew Bro…	The Nationals	<null>	https://twitter.com…
5	McMahon, New Sou…	https://www.ap…	Hon Chris Bow…	Australian La…	(02) 9604 0710	http://twitter.com/…
6	Canberra, Austra…	https://www.ap…	Ms Gai Brodtm…	Australian La…	(02) 6293 1344	<null>
7	McMillan, Victor…	https://www.ap…	Mr Russell Br…	Liberal Party…	(03) 5623 2064	<null>
8	Bendigo, Victoria	https://www.ap…	Ms Lisa Chest…	Australian La…	(03) 5443 9055	https://twitter.com…
9	Cunningham, New …	https://www.ap…	Hon Sharon Bi…	Australian La…	(02) 4228 5899	http://twitter.com/…
10	Wakefield, South…	https://www.ap…	Mr Nick Champ…	Australian La…	(08) 8284 2422	http://twitter.com/…

Table mp

	sa1_maincode_2016	ced_code_2017	ced_name_2017	state_code_2017	state_name_2017	area_albers_sqkm
1	11901135801	101	Banks	1	New South Wales	0.1146
2	11901135802	101	Banks	1	New South Wales	0.0722
3	11901135803	101	Banks	1	New South Wales	0.1597
4	11901135804	101	Banks	1	New South Wales	0.173
5	11901135805	101	Banks	1	New South Wales	0.108
6	11901135806	101	Banks	1	New South Wales	0.5865
7	11901135807	101	Banks	1	New South Wales	0.0845
8	11901135808	101	Banks	1	New South Wales	0.1662
9	11901135809	101	Banks	1	New South Wales	0.1406
10	11901135810	101	Banks	1	New South Wales	0.1577

Table 2

After all the data is in the database we can then run the 4 queries.